CSE 242 – Assignment 1

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Fall 2026

1 Linear Algebra

1.1 Q1

$A = \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}$.

$$AB = \begin{bmatrix} 3\cdot 1 + 4\cdot 0 & 3\cdot 2 + 4\cdot(-1) \\ 2\cdot 1 + 1\cdot 0 & 2\cdot 2 + 1\cdot(-1) \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}$$

$$BA = \begin{bmatrix} 1\cdot 3 + 2\cdot 2 & 1\cdot 4 + 2\cdot 1 \\ 0\cdot 3 + (-1)\cdot 2 & 0\cdot 4 + (-1)\cdot 1 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ -2 & -1 \end{bmatrix}$$

$$(A+B)^T = \begin{bmatrix} 4 & 6 \\ 2 & 0 \end{bmatrix}^T = \begin{bmatrix} 4 & 2 \\ 6 & 0 \end{bmatrix}$$

$$AB^T = \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} 3\cdot 1 + 4\cdot 2 & 3\cdot 0 + 4\cdot(-1) \\ 2\cdot 1 + 1\cdot 2 & 2\cdot 0 + 1\cdot(-1) \end{bmatrix} = \begin{bmatrix} 11 & -4 \\ 4 & -1 \end{bmatrix}$$

$$\mathrm{trace}(A) = 3 + 1 = 4, \qquad \mathrm{trace}(AB) = 3 + 3 = 6$$

1.2 Q2

$M = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$.

$$\det(M - \lambda I) = \det \begin{bmatrix} 2-\lambda & -1 \\ -1 & 2-\lambda \end{bmatrix} = (2-\lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3),$$

$$\det(M - \lambda I) = 0 \;\Rightarrow\; \lambda_1 = 1,\ \lambda_2 = 3.$$

Eigenvectors $v = (v_1, v_2)^T$ solve $(M - \lambda I)v = 0$:

$$\lambda_1 = 1: \quad \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0 \;\Rightarrow\; v_1 = v_2 \;\Rightarrow\; v = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\lambda_2 = 3: \quad \begin{bmatrix} -1 & -1 \\ -1 & -1 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0 \;\Rightarrow\; v_1 = -v_2 \;\Rightarrow\; v = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Check $Mv = \lambda v$:

$$M\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2-1 \\ -1+2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 1\cdot\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad M\begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 2+1 \\ -1-2 \end{bmatrix} = \begin{bmatrix} 3 \\ -3 \end{bmatrix} = 3\cdot\begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

1.3 Q3

Let $A \in \mathbb{R}^{m\times n}$, $B \in \mathbb{R}^{n\times m}$. Then

$$\operatorname{trace}(AB) = \sum_{i=1}^{m}(AB)_{ii} = \sum_{i=1}^{m}\sum_{j=1}^{n} A_{ij}B_{ji} = \sum_{j=1}^{n}\sum_{i=1}^{m} B_{ji}A_{ij} = \sum_{j=1}^{n}(BA)_{jj} = \operatorname{trace}(BA),$$

where the third equality exchanges the order of the two finite sums. □

1.4 Q4

$C = \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}$, $\det C = 4\cdot 6 - 7\cdot 2 = 10 \neq 0$, so C^{-1} exists. To compute it, row-reduce the augmented matrix $[\,C \mid I\,]$ until the left block becomes I; the right block is then C^{-1} (R_i denotes row i):

$$\left[\begin{array}{cc|cc} 4 & 7 & 1 & 0 \\ 2 & 6 & 0 & 1 \end{array}\right] \xrightarrow{R_1 \leftarrow R_1/4} \left[\begin{array}{cc|cc} 1 & \frac{7}{4} & \frac{1}{4} & 0 \\ 2 & 6 & 0 & 1 \end{array}\right] \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \left[\begin{array}{cc|cc} 1 & \frac{7}{4} & \frac{1}{4} & 0 \\ 0 & \frac{5}{2} & -\frac{1}{2} & 1 \end{array}\right]$$

$$\xrightarrow{R_2 \leftarrow \frac{2}{5}R_2} \left[\begin{array}{cc|cc} 1 & \frac{7}{4} & \frac{1}{4} & 0 \\ 0 & 1 & -\frac{1}{5} & \frac{2}{5} \end{array}\right] \xrightarrow{R_1 \leftarrow R_1 - \frac{7}{4}R_2} \left[\begin{array}{cc|cc} 1 & 0 & \frac{3}{5} & -\frac{7}{10} \\ 0 & 1 & -\frac{1}{5} & \frac{2}{5} \end{array}\right]$$

(last step: $\frac{1}{4} - \frac{7}{4}\cdot(-\frac{1}{5}) = \frac{1}{4} + \frac{7}{20} = \frac{3}{5}$, $0 - \frac{7}{4}\cdot\frac{2}{5} = -\frac{7}{10}$). Hence

$$C^{-1} = \begin{bmatrix} \frac{3}{5} & -\frac{7}{10} \\ -\frac{1}{5} & \frac{2}{5} \end{bmatrix} = \begin{bmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{bmatrix}.$$

2 Probability, MLE and MAP

2.1 Q1

$P(X = k) = c\,\dfrac{3^k}{k!}$, $k = 0, 1, 2, \ldots$

(a) The probabilities must sum to 1; with $\sum_{k\geq 0} x^k/k! = e^x$,

$$\sum_{k=0}^{\infty} c\,\frac{3^k}{k!} = c\,e^3 = 1 \;\Rightarrow\; c = e^{-3}.$$

(b)

$$\begin{aligned}
\mathbb{E}[X] &= \sum_{k=0}^{\infty} k\,e^{-3}\frac{3^k}{k!} = 3e^{-3}\sum_{k=1}^{\infty}\frac{3^{k-1}}{(k-1)!} = 3e^{-3}e^3 = 3 \\
\mathbb{E}[X(X-1)] &= \sum_{k=0}^{\infty} k(k-1)\,e^{-3}\frac{3^k}{k!} = 9e^{-3}\sum_{k=2}^{\infty}\frac{3^{k-2}}{(k-2)!} = 9e^{-3}e^3 = 9 \\
\mathbb{E}[X^2] &= \mathbb{E}[X(X-1)] + \mathbb{E}[X] = 9 + 3 = 12 \\
\operatorname{Var}(X) &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = 12 - 9 = 3 \qquad \text{(by Q6)}
\end{aligned}$$

2.2 Q2

$P(A) = 0.5$, $P(B) = 0.3$, $P(A \mid B) = 0.6$.

$$\text{(a)} \quad P(A \cap B) = P(A \mid B)\, P(B) = 0.6 \times 0.3 = 0.18$$

$$\text{(b)} \quad P(B \mid A) = \frac{P(A \cap B)}{P(A)} = \frac{0.18}{0.5} = 0.36$$

$$\text{(c)} \quad P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.5 + 0.3 - 0.18 = 0.62$$

2.3 Q3

Data H, H, T, H, T: 3 heads, 2 tails; each flip is Bernoulli(θ).
(a) Likelihood and log-likelihood:

$$L(\theta) = \theta \cdot \theta \cdot (1-\theta) \cdot \theta \cdot (1-\theta) = \theta^3 (1-\theta)^2, \qquad \ell(\theta) = \log L(\theta) = 3 \log \theta + 2 \log(1-\theta)$$

$$\ell'(\theta) = \frac{3}{\theta} - \frac{2}{1-\theta} = 0 \;\Rightarrow\; 3(1-\theta) = 2\theta \;\Rightarrow\; \hat{\theta}_{\text{MLE}} = \frac{3}{5}$$

$\ell''(\theta) = -3/\theta^2 - 2/(1-\theta)^2 < 0$ and $L(0) = L(1) = 0$, so this is the maximum.
(b) Posterior $\propto$ likelihood $\times$ prior:

$$P(\theta \mid \text{data}) \propto L(\theta) P(\theta) = \theta^3 (1-\theta)^2 \cdot 2\theta = 2\theta^4 (1-\theta)^2$$

$$\log P(\theta \mid \text{data}) = 4 \log \theta + 2 \log(1-\theta) + \text{const}$$

$$\frac{d}{d\theta} : \; \frac{4}{\theta} - \frac{2}{1-\theta} = 0 \;\Rightarrow\; 4(1-\theta) = 2\theta \;\Rightarrow\; \hat{\theta}_{\text{MAP}} = \frac{2}{3}$$

Second derivative $-4/\theta^2 - 2/(1-\theta)^2 < 0$ and the posterior is 0 at $\theta = 0, 1$, so this is the maximum.

2.4 Q4

For $x_1, \ldots, x_n$ i.i.d. $\mathcal{N}(\mu, \sigma^2)$ the log-likelihood is

$$\ell(\mu, \sigma^2) = \sum_{i=1}^{n} \log \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x_i - \mu)^2/(2\sigma^2)} = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i - \mu)^2.$$

Setting the partial derivatives to zero (data $\{1, 3, 5, 7\}$, $n = 4$):

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2} \sum_{i=1}^{n} (x_i - \mu) = 0 \qquad \Rightarrow\; \hat{\mu} = \frac{1}{n} \sum_{i=1}^{n} x_i = \frac{1+3+5+7}{4} = 4$$

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^{n} (x_i - \mu)^2 = 0 \quad \Rightarrow\; \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^{n} (x_i - \hat{\mu})^2 = \frac{(-3)^2 + (-1)^2 + 1^2 + 3^2}{4} = 5$$

2.5 Q5

$x \mid \mu \sim \mathcal{N}(\mu, \sigma^2)$, $\sigma^2 = 4$; $\mu \sim \mathcal{N}(\mu_0, \tau^2)$, $\mu_0 = 0$, $\tau^2 = 1$; $x = 5$. Posterior $\propto$ likelihood $\times$ prior:

$$\log P(\mu \mid x) = \log P(x \mid \mu) + \log P(\mu) + \text{const} = -\frac{(x-\mu)^2}{2\sigma^2} - \frac{(\mu - \mu_0)^2}{2\tau^2} + \text{const}$$

$$= -\frac{(5-\mu)^2}{8} - \frac{\mu^2}{2} + \text{const}$$

$$\frac{d}{d\mu} : \; \frac{5-\mu}{4} - \mu = 0 \;\Rightarrow\; 5 - \mu = 4\mu \;\Rightarrow\; \hat{\mu}_{\text{MAP}} = 1 \qquad \left(\tfrac{d^2}{d\mu^2} = -\tfrac{1}{4} - 1 < 0 \right)$$

2.6 Q6

Let $\mu = \mathbb{E}[X]$. By linearity of expectation,

$$\begin{aligned}\mathrm{Var}(X) &= \mathbb{E}[(X-\mu)^2] = \mathbb{E}[X^2 - 2\mu X + \mu^2] = \mathbb{E}[X^2] - 2\mu\,\mathbb{E}[X] + \mu^2 \\ &= \mathbb{E}[X^2] - 2\mu^2 + \mu^2 = \mathbb{E}[X^2] - (\mathbb{E}[X])^2. \qquad \square\end{aligned}$$

3 Principal Component Analysis

3.1 Q1

Let $X \in \mathbb{R}^d$ be a random vector with mean $\mu = \mathbb{E}[X]$ and covariance matrix $\Sigma = \mathbb{E}\big[(X-\mu)(X-\mu)^T\big]$, i.e. $\Sigma_{ij} = \mathrm{Cov}(X_i, X_j)$. For a unit vector $u \in \mathbb{R}^d$ let $Y = u^T X$. Then $\mathbb{E}[Y] = u^T\mathbb{E}[X] = u^T\mu$, and

$$\begin{aligned}\mathrm{Var}(Y) &= \mathbb{E}\big[(u^TX - u^T\mu)^2\big] = \mathbb{E}\big[(u^T(X-\mu))^2\big] \\ &= \mathbb{E}\big[u^T(X-\mu)(X-\mu)^Tu\big] && ((u^Tz)^2 = u^Tzz^Tu) \\ &= u^T\,\mathbb{E}\big[(X-\mu)(X-\mu)^T\big]\,u && (u \text{ constant, } \mathbb{E} \text{ linear}) \\ &= u^T\Sigma u. \qquad \square\end{aligned}$$

3.2 Q2

$n = 500$ samples drawn with NumPy (`multivariate_normal`, seed 0) from $\mathcal{N}(\mu, \Sigma)$ with

$$\mu = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \qquad \Sigma = \begin{bmatrix} 3 & 1.2 \\ 1.2 & 1 \end{bmatrix}, \qquad \text{correlation } \rho = \frac{\Sigma_{12}}{\sqrt{\Sigma_{11}\Sigma_{22}}} = \frac{1.2}{\sqrt{3 \cdot 1}} \approx 0.693.$$

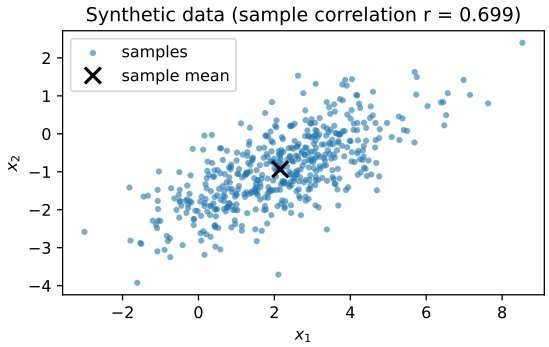


Figure 1: Synthetic samples and their sample mean.

Sample mean, covariance and correlation: $\bar{x} = \begin{bmatrix} 2.152 \\ -0.935 \end{bmatrix}$, $\hat{\Sigma} = \begin{bmatrix} 2.916 & 1.166 \\ 1.166 & 0.953 \end{bmatrix}$, $r = 0.699$. The cloud is an ellipse tilted with positive slope, i.e. x_1 and x_2 are positively correlated ($\Sigma_{12} > 0$); it is wider along x_1 because $\mathrm{Var}(x_1) = 3 > \mathrm{Var}(x_2) = 1$. With $r \approx 0.7$ the correlation is strong but not perfect: the points scatter around the line rather than on it.

3.3 Q3

1. Centre: with X the $n \times 2$ data matrix, $\bar{x} = \frac{1}{n}\sum_i x_i$ and $X_c = X - \mathbf{1}\bar{x}^T$.
2. Covariance: $\hat{\Sigma} = \frac{1}{n-1}X_c^TX_c$ (sample version of $\mathbb{E}[(X-\mu)(X-\mu)^T]$).

3. Eigendecomposition: $\hat{\Sigma} = V\Lambda V^T$ with `np.linalg.eigh`, eigenvalues sorted descending.

4. By Section 3.1, the variance of the data projected on a unit u is $u^T\hat{\Sigma}u$; for an eigenvector v_k it is $v_k^T\lambda_k v_k = \lambda_k$. So the columns of V (ordered by λ) are the principal components, and λ_k is the variance along each.

$$\hat{\Sigma} = \begin{bmatrix} 2.916 & 1.166 \\ 1.166 & 0.953 \end{bmatrix}, \qquad \lambda_1 = 3.458,\ \lambda_2 = 0.411, \qquad V = \begin{bmatrix} -0.907 & 0.422 \\ -0.422 & -0.907 \end{bmatrix}.$$

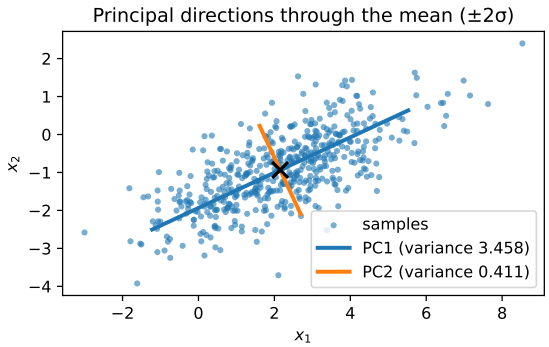


Figure 2: Principal directions through the sample mean; each segment spans $\pm 2\sqrt{\lambda_k}$.

PC1 lies along the long axis of the ellipse, PC2 is orthogonal along the short axis; the segment lengths $2\sqrt{\lambda_k}$ (3.7 vs 1.3) match the spread of the data in each direction.

3.4 Q4

Proportion of variance explained by component k:

$$f_k = \frac{\lambda_k}{\lambda_1 + \lambda_2}: \qquad f_1 = \frac{3.458}{3.869} = 0.894, \qquad f_2 = \frac{0.411}{3.869} = 0.106.$$

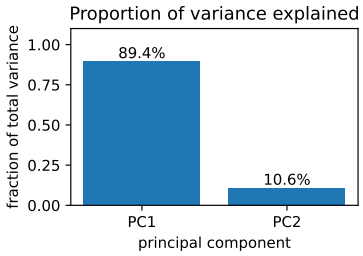


Figure 3: Proportion of variance explained by each component.

PC1 alone explains 89.4% of the variance, so projecting onto PC1 reduces the data from 2 to 1 dimension while keeping 89.4% of its variance. The reduced data is more compact and easier to visualise, and the dropped low-variance direction is often mostly noise. The trade-off: the 10.6% along PC2 is lost, including any signal it carried, because PCA ranks directions by variance, not by relevance to a task.